Artificial Intelligence

Basics of KNOWLEDGE & LOGICAL REASONING

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Reasoning: From Facts to Conclusions

The recurring question for the whole lecture: “Given what the agent knows, what can it safely conclude?”

01 KNOW

Facts + observations

02 REPRESENT

Translate into a formal language

03 INFER

Apply valid reasoning rules

04 ACT

Choose a justified action

Knowledge + logic → new conclusions → better decisions

REASONING

Where Does Logical Reasoning Appear?

The same ideas show up in policy systems, security rules, planning, and intelligent agents.

University rules

Prerequisites → permission to enroll

Cybersecurity

Login signals → suspicious event → MFA

Planning

State + constraints → legal next action

AI agents

Retrieved facts + tools → verified response

WARM-UP MYSTERY

Did It Rain?

Do not introduce the terminology yet. Let’s try to reason first.

CLUE 1

If it did not rain, Alex went to the outdoor festival.

CLUE 2

Alex went to either the outdoor festival or the indoor gaming event, but not both.

CLUE 3

Alex went to the indoor gaming event.

QUESTION: DID IT RAIN?

Click 1: Eliminate One Possibility

Use Clue 2 + Clue 3.

Known

Alex went to the indoor event.
Exactly one of the two events occurred.

New conclusion

Alex did NOT go to the outdoor festival.

This conclusion was not explicitly stored. It was inferred.

NEXT CLICK → use Clue 1

Click 2: Use the Rule Backward

If “no rain” implies outdoor festival, but outdoor festival is false...



This is the intuition behind Modus Tollens. We will formalize it later.

Existing knowledge → logical reasoning → NEW conclusion

Knowledge-Based Agent

An agent stores explicit knowledge and uses inference before acting.

ENVIRONMENT

What is happening?

PERCEPTS

What does the agent observe?

KNOWLEDGE BASE

What does it know?

INFERENCE

What can it figure out?

ACTION

What should it do?

KEY IDEA

Storing information is not enough. The agent must derive useful new information.

Question 1

Which system is primarily knowledge-based?

A. A neural network classifying handwritten digits

B. A clustering algorithm grouping customers

C. A university system applying prerequisite rules

D. A regression model predicting house prices

VOTE BEFORE THE NEXT CLICK

Question 1 — Answer

Which system is primarily knowledge-based?

A. A neural network classifying handwritten digits

B. A clustering algorithm grouping customers

C. A university system applying prerequisite rules

D. A regression model predicting house prices

WHY

A prerequisite system applies explicit facts and rules; its conclusions can be traced to those rules.

02

Propositional Logic

Give the agent a precise language for true/false statements.

Sentence → Proposition → Symbol

Formal logic removes ambiguity by giving each proposition a precise meaning.

Sentence

“The laboratory is open.”
A declarative statement.

Proposition

A statement that can be TRUE or FALSE.

Symbol

L = “The laboratory is open.”

NOT propositions

Commands, questions, and exclamations are not directly assigned a truth value.

Five Logical Connectives

These operators let us build larger sentences from simple propositions.

Negation

NOT P

$\neg P$

Conjunction

P AND Q

$P \wedge Q$

Disjunction

P OR Q

$P \vee Q$

Implication

IF P, THEN Q

$P \rightarrow Q$

Biconditional

P IFF Q

$P \leftrightarrow Q$

OR is inclusive unless stated otherwise: P ∨ Q allows one or both.

Truth Tables: Make Meaning Mechanical

Evaluate every possible assignment instead of relying on intuition.

P	Q	¬P	P ∧ Q	P ∨ Q
F	F	T	F	F
F	T	T	F	T
T	F	F	F	T
T	T	F	T	T

Classroom check

Why is P ∨ Q true in the last row?

Because OR means at least one is true; both may be true.

Implication: The Most Common Logic Trap

P → Q means: whenever P is true, Q must be true. It does not say P is the only cause of Q.

Rule

HighTemperature → CoolingFanOn

Observation

CoolingFanOn

Can we conclude HighTemperature?

NO — the fan could be on for another reason.

Question 2

Given P (safe system) → Q (no fire alarm) and Q is true, what can we conclude about P?

A. P must be true

B. P must be false

C. P may be true, but it is not guaranteed

D. The implication must be false

VOTE BEFORE THE NEXT CLICK

Question 2 — Answer

Given P → Q and Q is true, what can we conclude about P?

A. P must be true

B. P must be false

C. P may be true, but it is not guaranteed

D. The implication must be false

WHY

Inferring P from Q is the fallacy “affirming the consequent.” Q may have other causes.

Useful Equivalences

These transformations let an inference engine rewrite sentences into more useful forms.

Implication

$P \to Q \equiv \neg P \vee Q$

De Morgan 1

$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$

De Morgan 2

$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$

Biconditional

$P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)$

“Not both” ≠ “neither”: $\neg(H \wedge R) \equiv \neg H \vee \neg R$

03

Entailment & Model Checking

Does the conclusion have to be true in every world compatible with the KB?

Entailment: “Must Be True”

KB ⊨ α means α is true in every model in which the knowledge base is true.

Knowledge Base (KB)

Facts + observations + rules the agent accepts as true.

Query (α)

A sentence we ask the agent to evaluate.

$$KB \models \alpha \ ?$$

Not entailed does NOT automatically mean false — it may be unknown.

04

Inference Rules

Reason directly with sentences instead of enumerating every possible world.

04

Inference is just using what you already know to get new information. You have facts and rules, then you ask, "What follows from this?" For example,

if "If P then Q," and P is true, you infer Q. Inference is the process, entailment is the guarantee.

"Inference is how we derive a new conclusion; entailment is whether that conclusion is logically guaranteed.

04

Core Concepts of Inference

- **Evidence-based:** It relies on facts, clues, or observations rather than random guessing.
- **Prior Knowledge:** It uses what you already know to figure out what is not explicitly stated.
- **Logical Step:** It bridges the gap between raw data and a meaningful takeaway.

04

Common Examples

- **Daily Life:** Seeing dark, cloudy skies and inferring that it is going to rain.
- **Reading:** Noticing a character has wet eyes and a tissue, and inferring they are sad even if the text does not state it directly.
- **Statistics:** Using a small sample group to make a generalized conclusion about a larger population.

Two Core Inference Rules

Use them as a diagnostic for valid versus invalid reasoning.

Modus Ponens

$P \rightarrow Q$
P
———
$\therefore Q$

Example: HighTemp → FanOn; HighTemp; therefore FanOn.

Modus Tollens

$P \rightarrow Q$
$\neg Q$
———
$\therefore \neg P$

Example: ¬Rain → Outdoor; ¬Outdoor; therefore Rain.

INVALID: Q → therefore P | INVALID: ¬P → therefore ¬Q

Forward Chaining — Step 0

Start from facts and repeatedly fire rules whose premises are already known.

Rules

$A \rightarrow B$
$A \rightarrow C$
$B \wedge C \rightarrow D$
$D \wedge E \rightarrow Q$
$A \wedge D \rightarrow Q$

Known facts

$\{ A \}$

Why?

Only A is initially known.

Forward Chaining — Step 1

Start from facts and repeatedly fire rules whose premises are already known.

Rules

$A \to B$
$A \to C$
$B \wedge C \to D$
$D \wedge E \to Q$
$A \wedge D \to Q$

Known facts

{ A, B, C }

Newly inferred

B and C

Why?

$A \to B$ and $A \to C$ can both fire.

Forward Chaining — Step 2

Start from facts and repeatedly fire rules whose premises are already known.

Rules

$A \rightarrow B$
$A \rightarrow C$
$B \wedge C \rightarrow D$
$D \wedge E \rightarrow Q$
$A \wedge D \rightarrow Q$

Known facts

{ A, B, C, D }

Newly inferred

D

Why?

B and C are known, so $B \wedge C \rightarrow D$ fires.

Forward Chaining — Step 3

Start from facts and repeatedly fire rules whose premises are already known.

Rules

$A \rightarrow B$
$A \rightarrow C$
$B \land C \rightarrow D$
$D \land E \rightarrow Q$
$A \land D \rightarrow Q$

Known facts

{ A, B, C, D, Q }

Newly inferred

Q

Why?

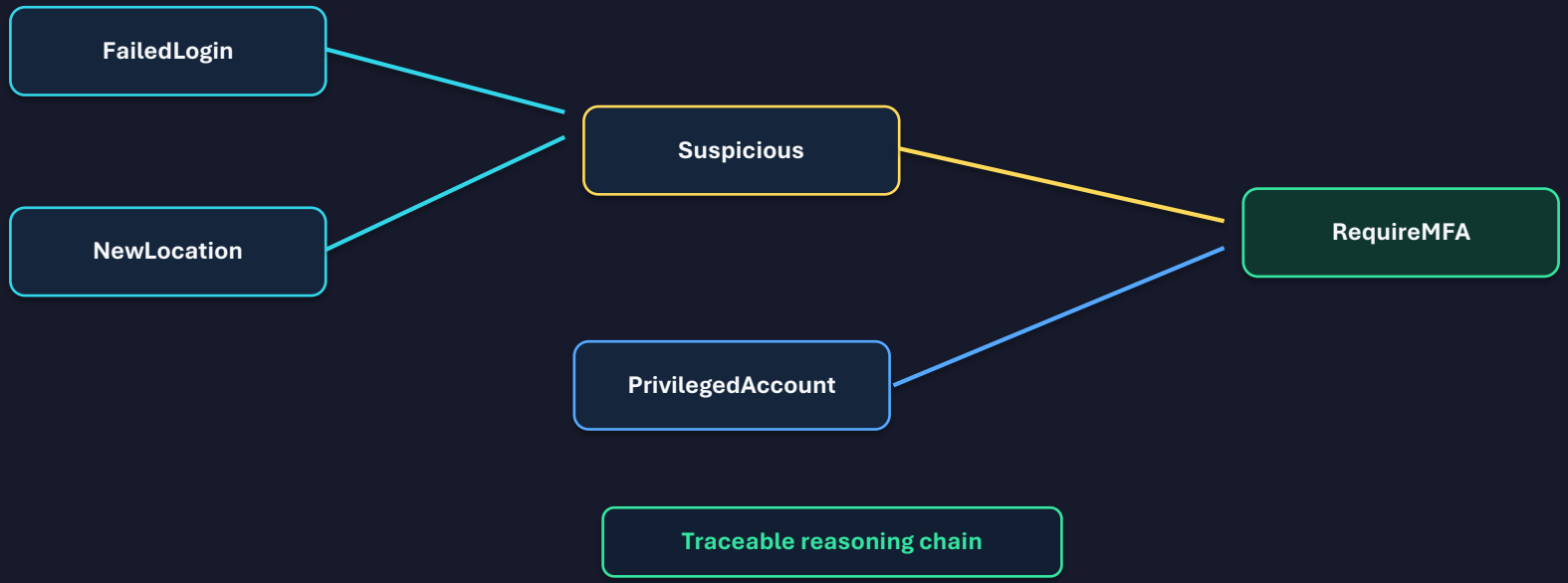
$D \land E \rightarrow Q$ cannot fire because E is unknown.
But $A \land D \rightarrow Q$ can fire.

Applied Example: Cybersecurity Rule Chain

A symbolic security layer can justify why it escalated an event.

FailedLogin ∧ NewLocation → Suspicious

Suspicious ∧ PrivilegedAccount → RequireMFA



Proof by Contradiction

To prove KB ⊨ α, add ¬α. If resolution produces the empty clause, α is entailed.

Assume

¬A

Resolve with ¬C

¬B

Resolve with A∨B

A

Resolve A with ¬A

□

Empty clause □

Contradiction reached → the opposite assumption cannot be true → KB entails A.

05

First-Order Logic

Represent objects, relationships, and reusable rules — not only whole facts.

07

Reasoning in Large Language Models

How does modern LLM reasoning relate to the symbolic logic we just studied?

SEARCH CONCEPT CHECK

Question 1

Which statement best defines uninformed search?

A. Search that does not use a heuristic

B. Search that always uses h(n)

C. Search used only for games

D. Search that finds the goal in one step

VOTE BEFORE THE NEXT CLICK

SEARCH CONCEPT CHECK

Question 1 — Answer

Which statement best defines uninformed search?

A. Search that does not use a heuristic

B. Search that always uses h(n)

C. Search used only for games

D. Search that finds the goal in one step

WHY

Uninformed search explores without heuristic guidance.

SEARCH CONCEPT CHECK

Question 2

What does the heuristic h(n) represent?

A. Exact cost from the start to n

B. Number of nodes already explored

C. Estimated remaining cost from n to the goal

D. Utility value of an opponent

VOTE BEFORE THE NEXT CLICK

SEARCH CONCEPT CHECK

Question 2 — Answer

What does the heuristic h(n) represent?

A. Exact cost from the start to n

B. Number of nodes already explored

C. Estimated remaining cost from n to the goal

D. Utility value of an opponent

WHY

h(n) estimates the remaining cost from the current node to the goal.

Question 3

What does Greedy Best-First Search choose first?

A. The node with the smallest g(n) + h(n)

B. The node with the smallest h(n)

C. The node with the largest h(n)

D. The node with the smallest g(n)

VOTE BEFORE THE NEXT CLICK

Question 3 — Answer

What does Greedy Best-First Search choose first?

A. The node with the smallest g(n) + h(n)

B. The node with the smallest h(n)

C. The node with the largest h(n)

D. The node with the smallest g(n)

WHY

Greedy search follows the heuristic only and chooses the smallest h(n).

SEARCH CONCEPT CHECK

Question 4

In A* search, what does g(n) mean?

A. Estimated remaining cost to the goal

B. The heuristic value

C. The opponent utility

D. Actual cost from the start to node n

VOTE BEFORE THE NEXT CLICK

SEARCH CONCEPT CHECK

Question 4 — Answer

In A* search, what does g(n) mean?

A. Estimated remaining cost to the goal

B. The heuristic value

C. The opponent utility

D. Actual cost from the start to node n

WHY

g(n) is the actual path cost already paid from the start to node n.

Question 5

Which formula is used by A* search?

A. f(n) = h(n)

B. f(n) = g(n)

C. f(n) = g(n) + h(n)

D. f(n) = g(n) - h(n)

VOTE BEFORE THE NEXT CLICK

Question 5 — Answer

Which formula is used by A* search?

A. f(n) = h(n)

B. f(n) = g(n)

C. f(n) = g(n) + h(n)

D. f(n) = g(n) - h(n)

WHY

A* combines the cost so far with the estimated remaining cost.

SEARCH CONCEPT CHECK

Question 6

What is the main difference between informed and uninformed search?

A. Informed search is always slower

B. Informed search uses heuristic guidance

C. Uninformed search always uses an opponent

D. There is no difference

VOTE BEFORE THE NEXT CLICK

SEARCH CONCEPT CHECK

Question 6 — Answer

What is the main difference between informed and uninformed search?

A. Informed search is always slower

B. Informed search uses heuristic guidance

C. Uninformed search always uses an opponent

D. There is no difference

WHY

Informed search uses h(n) to guide exploration toward the goal.

SEARCH CONCEPT CHECK

Question 7

When is search called adversarial?

A. When another agent or opponent actively affects the outcome

B. When there is no goal

C. When there are no possible actions

D. When only path cost matters

VOTE BEFORE THE NEXT CLICK

SEARCH CONCEPT CHECK

Question 7 — Answer

When is search called adversarial?

A. When another agent or opponent actively affects the outcome

B. When there is no goal

C. When there are no possible actions

D. When only path cost matters

WHY

Adversarial search plans while considering actions chosen by an opponent.

SEARCH CONCEPT CHECK

Question 8

In Minimax, what does the MAX player try to do?

A. Choose the lowest heuristic

B. Choose the lowest path cost only

C. Choose a random utility value

D. Choose the highest utility value

VOTE BEFORE THE NEXT CLICK

Question 8 — Answer

In Minimax, what does the MAX player try to do?

A. Choose the lowest heuristic

B. Choose the lowest path cost only

C. Choose a random utility value

D. Choose the highest utility value

WHY

MAX selects the move that gives the highest utility value.

Question 9

In Minimax, what does the MIN player try to do?

A. Choose the highest utility value

B. Choose the lowest utility value

C. Choose the shortest path

D. Choose the smallest heuristic h(n)

VOTE BEFORE THE NEXT CLICK

Question 9 — Answer

In Minimax, what does the MIN player try to do?

A. Choose the highest utility value

B. Choose the lowest utility value

C. Choose the shortest path

D. Choose the smallest heuristic h(n)

WHY

MIN represents the opponent trying to reduce MAX’s utility.

SEARCH CONCEPT CHECK

Question 10

What is the difference between a plan and a strategy in adversarial search?

A. A plan includes every opponent response; a strategy is one path

B. They mean exactly the same thing

C. A strategy includes responses to possible opponent moves

D. A strategy ignores the opponent

VOTE BEFORE THE NEXT CLICK

Question 10 — Answer

What is the difference between a plan and a strategy in adversarial search?

A. A plan includes every opponent response; a strategy is one path

B. They mean exactly the same thing

C. A strategy includes responses to possible opponent moves

D. A strategy ignores the opponent

WHY

A strategy specifies what to do for different possible opponent responses.

KNOWLEDGE CONCEPT CHECK

Question 11

What best describes a knowledge-based agent?

A. A system that only predicts patterns

B. A system that stores explicit knowledge and uses inference

C. A system that has no rules

D. A system that must always be a neural network

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 11 — Answer

What best describes a knowledge-based agent?

A. A system that only predicts patterns

B. A system that stores explicit knowledge and uses inference

C. A system that has no rules

D. A system that must always be a neural network

WHY

A knowledge-based agent stores facts or rules and derives useful conclusions.

KNOWLEDGE CONCEPT CHECK

Question 12

What is a proposition in propositional logic?

A. A statement that can be true or false

B. A question

C. A command

D. An exclamation

VOTE BEFORE THE NEXT CLICK

Question 12 — Answer

What is a proposition in propositional logic?

A. A statement that can be true or false

B. A question

C. A command

D. An exclamation

WHY

A proposition is a declarative statement with a truth value.

KNOWLEDGE CONCEPT CHECK

Question 13

What does the expression P ∧ Q mean?

A. P OR Q

B. NOT P

C. IF P, THEN Q

D. P AND Q

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 13 — Answer

What does the expression P ∧ Q mean?

A. P OR Q

B. NOT P

C. IF P, THEN Q

D. P AND Q

WHY

The symbol ∧ is conjunction, meaning AND.

KNOWLEDGE CONCEPT CHECK

Question 14

What does the implication P → Q mean?

A. P AND Q

B. P if and only if Q

C. If P, then Q

D. NOT Q

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 14 — Answer

What does the implication P → Q mean?

A. P AND Q

B. P if and only if Q

C. If P, then Q

D. NOT Q

WHY

P → Q means that when P is true, Q must be true.

KNOWLEDGE CONCEPT CHECK

Question 15

What is inference?

A. Storing facts without using them

B. Deriving new conclusions from known facts and rules

C. Removing all rules from the knowledge base

D. Guessing without evidence

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 15 — Answer

What is inference?

A. Storing facts without using them

B. Deriving new conclusions from known facts and rules

C. Removing all rules from the knowledge base

D. Guessing without evidence

WHY

Inference uses existing knowledge to derive new information.

KNOWLEDGE CONCEPT CHECK

Question 16

What does KB ⊨ α mean?

A. α is true in every model where the knowledge base is true

B. α is probably true

C. α must already be written as a fact in the KB

D. The knowledge base contains no rules

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 16 — Answer

What does KB ⊨ α mean?

A. α is true in every model where the knowledge base is true

B. α is probably true

C. α must already be written as a fact in the KB

D. The knowledge base contains no rules

WHY

Entailment means the conclusion must be true whenever the KB is true.

KNOWLEDGE CONCEPT CHECK

Question 17

Which pattern is Modus Ponens?

A. $P \rightarrow Q$ and Q, therefore P

B. $\neg P$, therefore $\neg Q$

C. $P \rightarrow Q$ and P, therefore Q

D. $P \vee Q$, therefore P

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 17 — Answer

Which pattern is Modus Ponens?

A. P → Q and Q, therefore P

B. ¬P, therefore ¬Q

C. P → Q and P, therefore Q

D. P ∨ Q, therefore P

WHY

Modus Ponens uses P → Q together with P to infer Q.

KNOWLEDGE CONCEPT CHECK

Question 18

What does forward chaining start from?

A. Only the final goal

B. An empty knowledge base

C. The opponent utility

D. Known facts

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 18 — Answer

What does forward chaining start from?

A. Only the final goal

B. An empty knowledge base

C. The opponent utility

D. Known facts

WHY

Forward chaining begins with known facts and fires applicable rules.

KNOWLEDGE CONCEPT CHECK

Question 19

What is the basic idea of resolution?

A. Compare two path costs

B. Combine clauses that contain complementary literals

C. Estimate h(n)

D. Predict the next token

VOTE BEFORE THE NEXT CLICK

KNOWLEDGE CONCEPT CHECK

Question 19 — Answer

What is the basic idea of resolution?

A. Compare two path costs

B. Combine clauses that contain complementary literals

C. Estimate h(n)

D. Predict the next token

WHY

Resolution removes complementary literals to derive a new clause.

KNOWLEDGE CONCEPT CHECK

Question 20

What is the basic structure of Conjunctive Normal Form (CNF)?

A. An AND of OR-clauses

B. An OR of AND-clauses only

C. One proposition with no operators

D. A heuristic formula

VOTE BEFORE THE NEXT CLICK

Question 20 — Answer

What is the basic structure of Conjunctive Normal Form (CNF)?

A. An AND of OR-clauses

B. An OR of AND-clauses only

C. One proposition with no operators

D. A heuristic formula

WHY

CNF represents a sentence as a conjunction (AND) of disjunctions (OR-clauses).